

256B on the C64 for 8K

Nick Montfort, 2019

Abstract

- Art Exhibition for Creativity & Cognition / Design Interactive Systems 2019
- Contribution: a 256B exe written in 6502 asm for the Commodore 64, displayed on the auditorium's 8K display.
- The code was executed on an original C64, not an emulator!



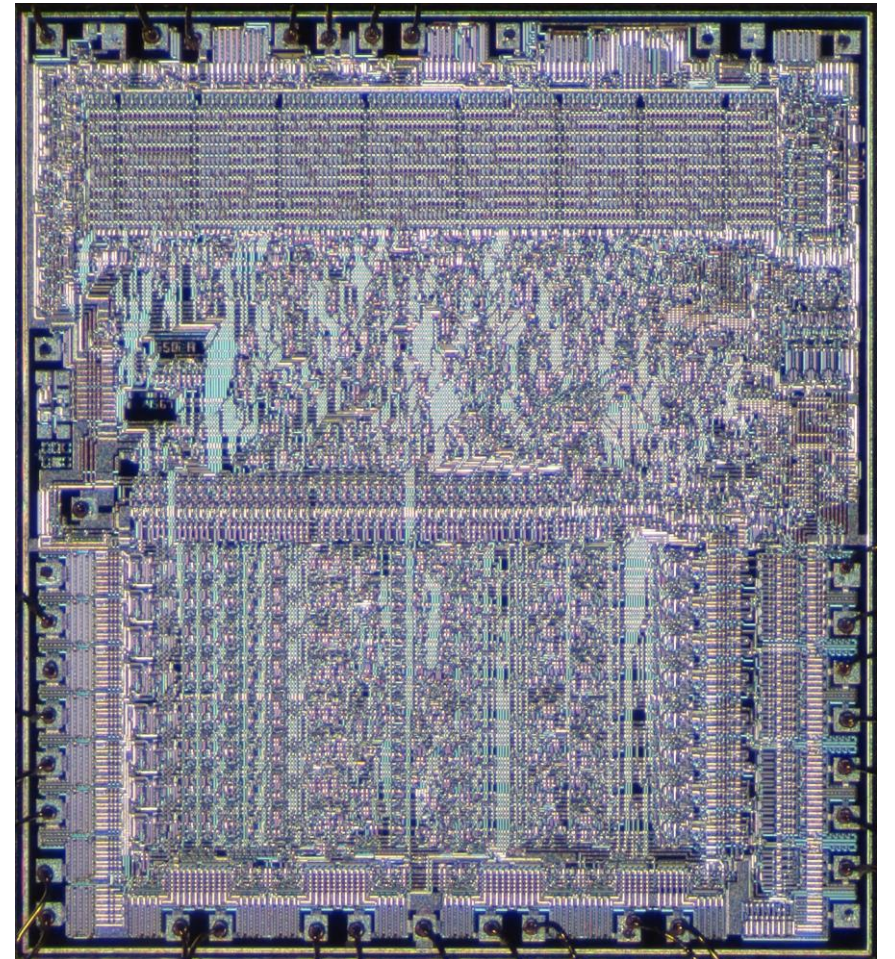
Introduction

- Project is a non-interactive, text-mode program running on the C64.
- Paper is a discussion of how the project relates to the author's works and interests.
- *The program itself will be made available for anyone to download and run on their C64 or in an emulator...*



The 6502

- 8-bit CPU (16-bit addressing) launched in 1975 by MOS Technology.
- 1 MHz – 3 MHz!
- 3 general purpose registers
- 56 instructions with various addressing modes (absolute, indirect, zero page)



1. Computational Concrete Poetry

- **Concrete poetry:** *an arrangement of linguistic elements in which the typographical effect is more important in conveying meaning than verbal significance* (Wikipedia).
- It is the visual appearance and arrangement of the words on the page, rather than the words themselves and what they mean.

Lord, who createdst man in wealth and flore,
Though foolishly he lost the fame,
Decaying more and more,
Till he became
Most poore:
With thee
O let me rife
As larks, harmoniously,
And sing this day thy victories:
Then shall the fall further the flight in me.
My tender age in sorrow did beginne
And fill with sicknesses and shame
Thou didst so punish finne,
That I became
Most thine.
With thee
Let me combine,
And feel this day thy victorie:
For, if I imp my wing on thine,
Affliction shall advance the flight in mee.

George Herbert's "Easter Wings", 1633

- **Computational poetry:** *the use of computer programs, algorithms, and code to generate, structure, or experience poetry.*
- Ex: Program generates poetry from a fixed set of words while following a grammatical structure
- Ex: « Lost poetry », extracting phrases with a poetic nature from larger blocks of text.



Fig 3 of *Computational Poetry is Lost Poetry* (Kreminski, 2024)

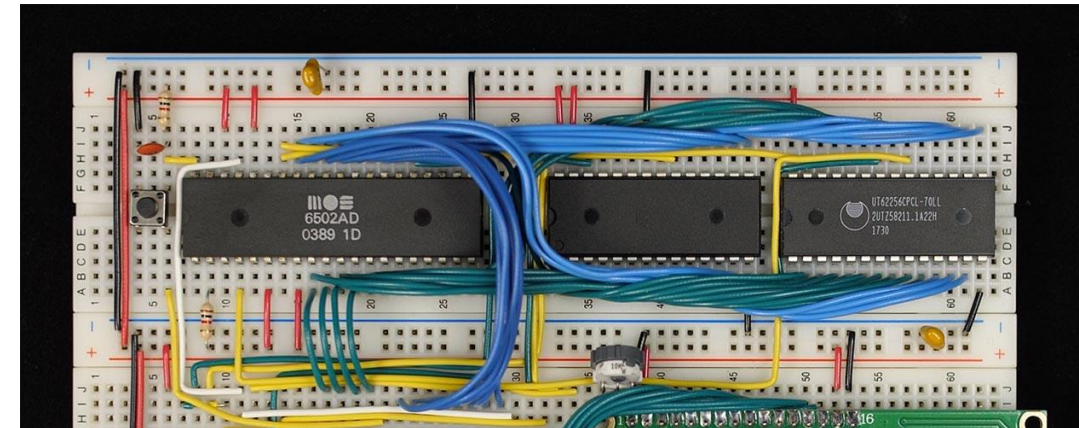
2. Retrocomputing

- Intentional use of old technology over modern equivalents
- Examples:
 - Audio recording on magnetic tapes
 - Chemical film photography
 - Sound artists using cassettes



2. Retrocomputing

- 6502 is a very old, retro CPU
- Used in many classic systems: Apple I & II, Atari, NES, BBC Micro, other Commodore systems.
- Very active retro-dev community even until today!
- Author's previous works: assembly and BASIC programs on the C64



3. Sizecoding

- Intentionally small programs (4K, 8K, etc)
- Constrained composition leads to an interesting creative tension
- Squeeze as much “wow” into as little space as possible

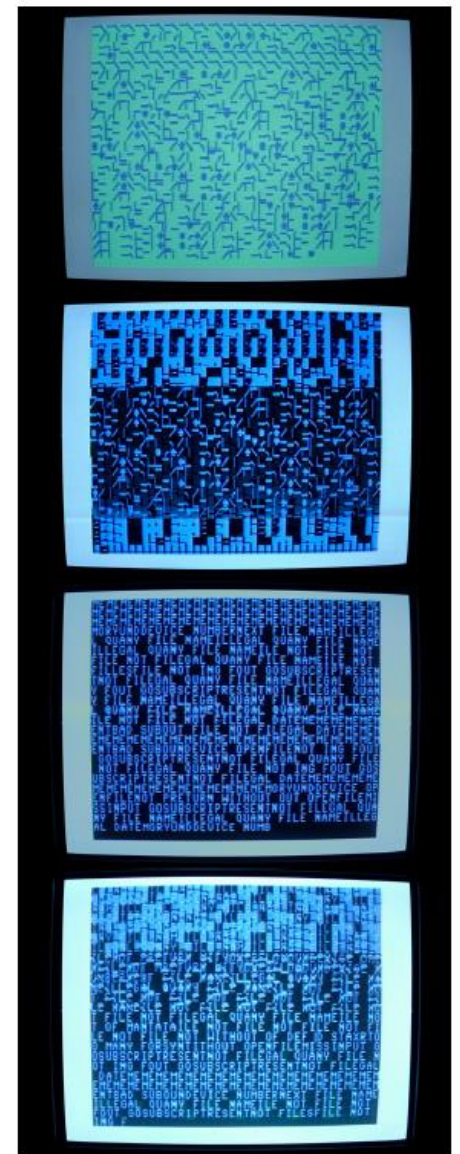


Figure 1: *Traumaphore* (2017), a 64-byte C64 program I wrote in 6502 assembly. Among other things it generates text using a bigram per-character Markov chain using BASIC error messages from ROM.

4. The Demoscene

- Directly links retro-computing and sizecoding.
- Programmers make size-constrained programs, “demo”s, short audio-visual presentations
- “64K intros”
- The sound is part of the size constraint!
- All sorts of algorithmic “tricks” to avoid using external assets/data.



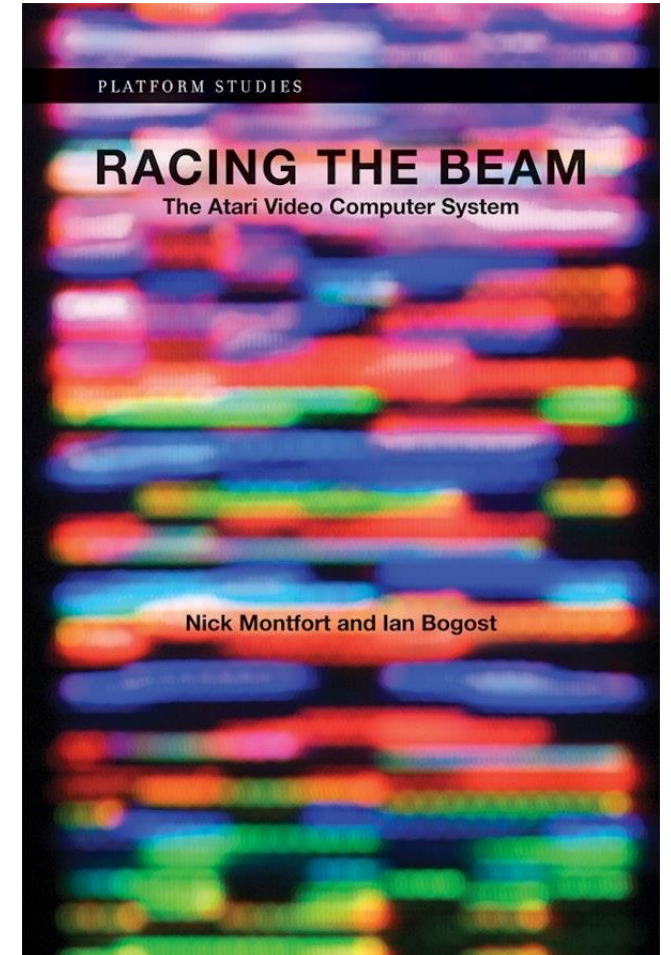
The Assembly 2004 – a Finnish demoscene & LAN party

Please the Cookie Thing – Aardbei (2000)



5. Platform Studies and Practices

- Editor of the *Platform Studies* book series (MIT Press).
- Series investigates various computing platforms, rather than the programs.
- Focus is on the connections between the platform and the surrounding cultural / societal environment.



Polytropon

an Applesoft BASIC Poem for Caroline Bergvall

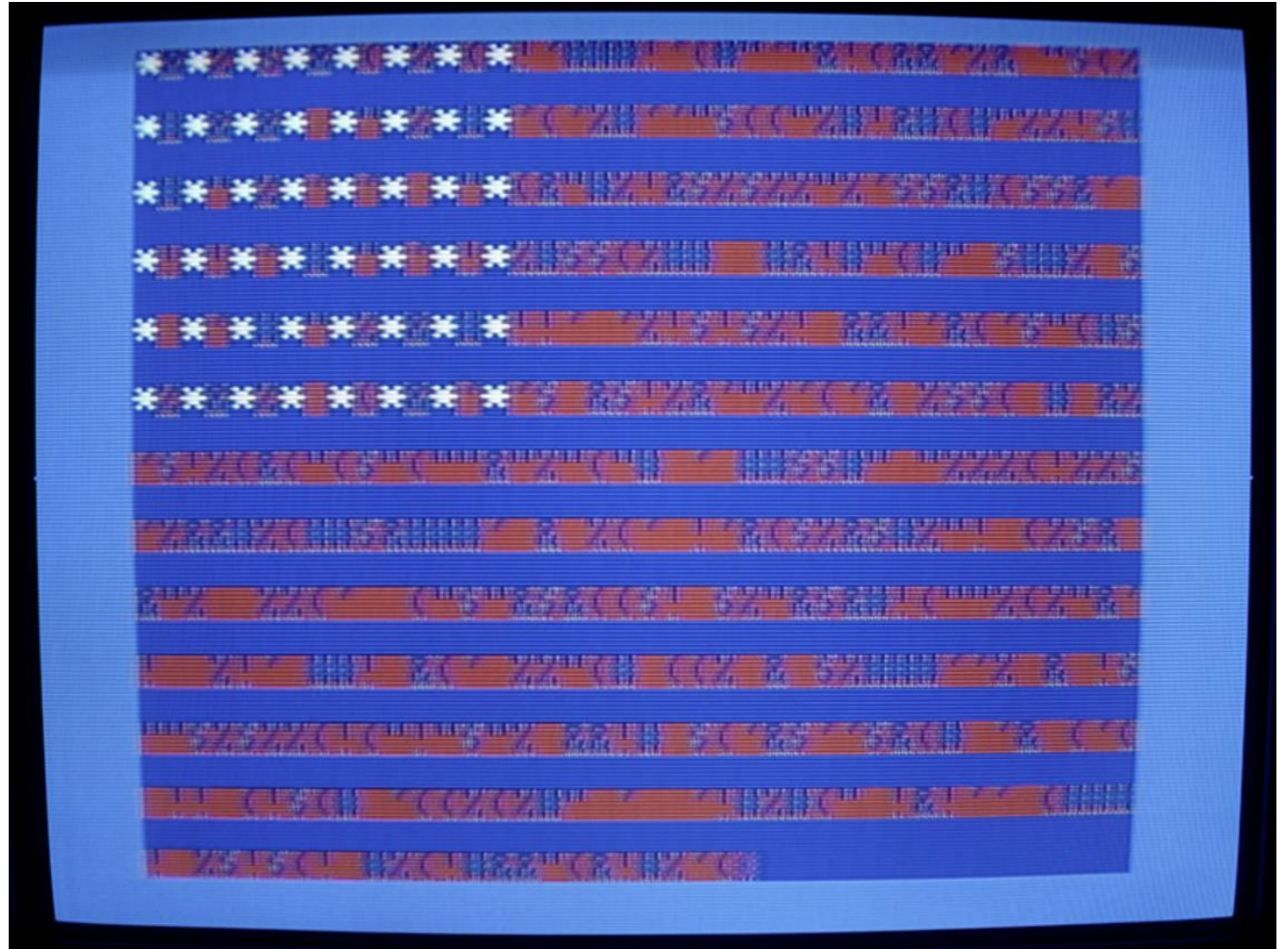
```
10 FOR I = 0 TO 9
20 READ A$(I)
30 NEXT I
40 FOR J = 0 TO 999
50 NEXT J
60 PRINT
70 PRINT A$(RND(1) * 10)
80 GOTO 40
90 DATA "Gone sing 'bout"
100 DATA "MY TALE BEGINS"
110 DATA "O DIVINE POSEY"
120 DATA "Remind us, Muse"
130 DATA "Sing in me, Muse"
140 DATA "Sing to me"
150 DATA "SPEAK, MEMORY-"
160 DATA "Tell me, Muse"
170 DATA "TELL ME, O MUSE"
180 DATA "Tell me"
```

The functioning of this program can be described pretty easily. It displays one of the ten lines represented in 90–180 at random, pauses briefly, and then does that again ... limitlessly. This program/poem is based on Bergvall's magnificent "Via," [11](#) a work that consists of 48 opening tercets from Dante's *Inferno*, in 48 different English translations, arranged in alphabetical order.

Flag

a Commodore 64 BASIC v2 poem for Jasper Johns

```
10 PRINT CHR$(18)CHR$(28);  
20 FOR I = 0 TO 39  
30 PRINT CHR$(32 + RND(1) * 9);  
40 IF I > 8 GOTO 70  
50 POKE 55696 + 2 * I, 1  
60 POKE 1440 + 2 * I, 42  
70 NEXT I  
80 PRINT  
90 GOTO 10
```



Conclusion

- 5 (quite different!) practices: artistic, literary, and computing.
- Interesting discussion of how the boundaries/lines between them get blurred.
- This project in particular:
 - « ...weaves together 1980s “home computing” with late 2010s video technology, the international cross-border connections made in the demoscene, and the intersection of five different types of practice and context ».

Questions?

References

- *256B on the C64 for 8K*, (Montfort, 2019),
https://nickm.com/articles/Montfort_256B_C64_8K.pdf
- https://en.wikipedia.org/wiki/Concrete_poetry
- <https://mitpress.mit.edu/series/platform-studies/>
- *Computational Poetry is Lost Poetry*, (Kreminski, 2024),
<https://dl.acm.org/doi/fullHtml/10.1145/3686169.3686179>
- https://en.wikipedia.org/wiki/MOS_Technology_6502
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